It is mandatory to have the table schema defined and developed in a database before retrieving or persisting data. But we can prime Elasticsearch with documents without defining a schema for our data model. This schema-free feature helps developers get up and running with the system from day one. However, best practice is to develop a schema up front rather than letting Elasticsearch define it for us, unless our requirements do not need one.

Elasticsearch expects us to provide clues about how it should treat a field when indexing data. These clues are either provided by us in the form of a schema definition while creating the index or implicitly derived by the engine if we allow it to do so. This process of creating the schema definition is called *mapping*.

Mapping allows Elasticsearch to understand the shape of the data so it can apply a set of predefined rules on fields before indexing them. Elasticsearch also consults the manual of mapping rules to apply full-text rules on text fields. Structured fields (exact values, like numbers or dates) have a separate set of instructions enabling them to be part of aggregations and sorting and filtering functions, in addition to being available for general searches.

In this chapter, we set the context for using mapping schemas, explore the mapping process, and work with data types, looking at how to define them using the mapping APIs. Data that is indexed for Elasticsearch has a definite shape and form. Meticulous shaping of data lets Elasticsearch do a faultless analysis, providing the end user with precise results. This chapter discusses the treatment of data in Elasticsearch and how mapping schemas help us avoid hindrances and obtain accurate searches.

100 m run or 400 m hurdles?

This chapter deals with dozens of hands-on examples around both core and advanced data types. While I advise you to read about them in the given order, if you are just starting with Elasticsearch and wish to focus on the beginner elements, you can skip section 4.6 and revisit it when you are more confident and want to learn more.

If all you want is a 100 m run rather than 400 m hurdles, read the chapter up to the core data types (section 4.4) and then feel free to jump to the next chapter.

4.1 Overview of mapping

Mapping is a process of defining and developing a schema definition representing a document's data fields and their associated data types. Mapping tells the engine the shape and form of the data that's being indexed. Because Elasticsearch is a document-oriented database, it expects a single mapping definition per index. Every field is treated according to the mapping rule. For example, a string field is treated as a text field, a number field is stored as an integer, a date field is indexed as a date to allow for date-related operations, and so on. Accurate and error-free mapping allows Elasticsearch to analyze data faultlessly, aiding search-related functionalities, sorting, filtering, and aggregation.